



IOT-BASED INTRUDER ALARM SYSTEM



A PROJECT REPORT

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In partial fulfilment for the award of the degree

of

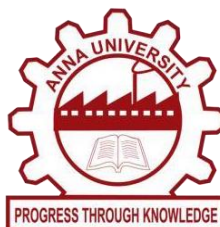
BACHELOR OF ENGINEERING

in

**COMPUTER SCIENCE AND ENGINEERING (INTERNET OF THINGS
AND CYBER SECURITY INCLUDING
BLOCKCHAIN TECHNOLOGY)**

SNS COLLEGE OF ENGINEERING

COIMBATORE - 641107



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BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

I wish to express sincere thanks to our management for providing us with the facilities to carry out this project work.

I'm greatly indebted to our Director **Dr.V.P.Arunachalam, Ph.D** and our Principal **Dr.S.Charles, Ph.D** for his continuous evaluation and critical suggestion given to complete the project work successfully.

I'm highly grateful to **Dr.P.Gnanasundari, M.E., Ph.D.**, Head of the Department of Electronics and Communication Engineering for her valuable suggestions and monitoring during the course of our project work.

I'm highly grateful to our Project Guide **Dr.P.Gnanasundari, M.E., Ph.D**, Assistant Professor, Department of Electronics and Communication Engineering for her valuable suggestions during the course of our project work.

I express my heartfelt and deep sense of gratitude to our Project coordinator **Dr.P.Gnanasundari, M.E., Ph.D**, Assistant Professor, Department of Electronics and Communication Engineering for her valuable advice and guidance. Her positive approach had offered incessant help in all possible ways from the beginning.

By this, I express my heartfelt sense of gratitude and thanks to our beloved family and friends who have all helped in collecting sources and materials needed for this project and for their support during our course and project work.

INDEX

CH.NO	TITLE	PG.NO
1	ABSTRACT	1
2	INTRODUCTION	2
3	PROBLEM STATEMENT	3
4	LITERATURE SURVEY	5
5	EXISTING SYSTEM	7
6	PROPOSED SYSTEM	8
7	BLOCL DIAGRAM	9
8	HARDWARE REQUIREMENTS	10
9	SOFTWARE REQUIREMENTS	15
10	WORKING PRINCIPLE	17
11	SOURCE CODE	19
12	RESULT	23
13	CONCLUSION	24
14	REFERENCE	25

ABSTRACT

The growing prevalence of burglary and unauthorized access necessitates more robust security solutions for residential and commercial properties. Traditional security systems, while providing a basic level of protection, often lack real-time monitoring and timely notification capabilities. This paper proposes an IoT-based intruder alarm system to address these limitations. By leveraging modern sensor technology and real-time communication, this system offers significant advancements in property security.

The core of the system revolves around the ESP32 microcontroller, which integrates seamlessly with Passive Infrared (PIR) sensors for motion detection and a buzzer for immediate alerts. The system is further augmented by a mobile application, enabling remote monitoring and control of the alarm system. Through this application, users receive real-time alerts, view the status of their premises, and manage the alarm system from anywhere with internet connectivity.

The proposed system provides real-time alerts upon suspicious activity detection, utilizes diverse sensors for proactive intrusion identification, and empowers users with remote monitoring capabilities for enhanced situational awareness.

Overall, the IoT-based Intruder Alarm System represents a comprehensive and efficient solution for property security, empowering users to protect their assets with ease and confidence.

CHAPTER 1

INTRODUCTON

The IoT-based Intruder Alarm System project, leveraging the capabilities of the ESP32 microcontroller, represents a significant advancement in the field of security and surveillance systems. The core objective of this project was to develop a robust and user-friendly platform that utilizes IoT technology to detect and alert the users. This endeavour not only aimed to bridge the gap between advanced IoT technologies and practical user applications but also sought to explore the innovative possibilities of IoT in enhancing security measures.

The integration of the ESP32 microcontroller was pivotal, granting access to a suite of advanced features suitable for IoT applications, including Wi-Fi connectivity and efficient processing power. The system incorporated Passive Infrared (PIR) sensors to detect motion, a buzzer for immediate auditory alerts, and a mobile application for remote monitoring and control. This project was designed to cater to a diverse audience, ranging from homeowners seeking enhanced security to technologists and educators exploring the capabilities of IoT in real-world applications. By focusing on user-friendly interface design and seamless backend integration, the project aimed to demystify IoT technology, making it accessible and usable for people without technical expertise in electronics or programming.

Through this initiative, the project highlighted the transformative potential of IoT in augmenting security systems and opened avenues for further exploration in the intersection of technology, home security, and human-computer interaction.

CHAPTER 2

PROBLEM STATEMENT

The rise in burglary and unauthorized entry incidents in residential and commercial properties has highlighted the critical need for effective security measures. Traditional security systems, while providing a certain level of protection, often fall short in several key areas. These conventional systems typically lack real-time monitoring capabilities, fail to provide timely alerts to property owners or authorities, and do not offer remote access for immediate intervention. Consequently, property owners are often left unaware of intrusions until it is too late to take preventative actions, leading to significant financial and emotional losses.

Traditional security systems have several significant limitations:

- 1. Limited Monitoring:** These systems typically lack real-time monitoring capabilities, hindering homeowners and business owners from being promptly informed of potential security breaches. This delay in awareness can be detrimental, allowing intruders valuable time to operate before authorities are notified.
- 2. Reactive Approach:** Traditional systems often rely on alarms triggered after an intrusion has already occurred. This reactive approach fails to provide a proactive defense mechanism, leaving properties vulnerable during the initial stages of an attempted break-in.
- 3. Limited Detection Methods:** Conventional systems often rely solely on door and window contacts, which can be bypassed by determined intruders. The lack of diverse sensor technology limits the system's ability to detect more sophisticated intrusion attempts.

Therefore, there is a pressing demand for an advanced intruder alarm system that leverages modern sensor technology to detect unauthorized access and promptly notify stakeholders. Such a system should not only enhance the detection capabilities but also

ensure immediate communication of security breaches to users, regardless of their location. This can be achieved by integrating Internet of Things (IoT) technologies, which offer superior connectivity and control.

The proposed solution involves the development of an IoT-based intruder alarm system using the ESP32 microcontroller, Passive Infrared (PIR) sensors, and a buzzer for immediate alerts. Additionally, the system will include a mobile application for remote monitoring and control, ensuring users can manage and respond to security alerts in real time. The ESP32's built-in Wi-Fi capability will enable seamless communication between the sensors and the user's mobile device, providing instant notifications and enhancing the overall security of the premises.

By addressing the limitations of traditional security systems, this IoT-based intruder alarm system aims to provide a comprehensive, user-friendly, and efficient security solution. This system will empower property owners with the tools needed to protect their assets more effectively, offering peace of mind through enhanced vigilance and prompt responsiveness. This advanced system will provide:

- 1. Real-time Monitoring and Alerts:** Property owners and relevant authorities will receive immediate notifications upon any suspicious activity detected by the system's sensors.
- 2. Proactive Defense:** By employing a wider range of sensors, including motion detectors, glass break sensors, and even video surveillance, the system can identify potential intrusions at their earliest stages, deterring criminal activity.
- 3. Enhanced Situational Awareness:** Remote monitoring capabilities will allow property owners to view the situation in real-time, enabling them to take informed actions or contact emergency services.

CHAPTER 3

LITERATURE SURVEY

LITERATURE SURVEY NO 1:

Title: " The Evolution of Intruder Alarm Systems: From Mechanical to Digital "

Authors: John M. Smith

Synopsis: This paper provides a historical overview of intruder alarm systems, tracing their development from mechanical devices in ancient times to sophisticated digital systems in the modern era.

LITERATURE SURVEY NO 2:

Title: "Sensor Technologies in Modern Intruder Alarm Systems"

Authors: Tero Karras, Timo Aila, Samuli Laine, Jaakko Lehtinen

Synopsis: The authors present a new training methodology for GANs, where the network grows progressively, enhancing the quality and stability of the generated images.

LITERATURE SURVEY NO 3:

Title: "BigGAN: Large Scale GAN Training for High Fidelity Natural Image Synthesis"

Authors: Andrew Brock, Jeff Donahue, Karen Simonyan

Synopsis: This paper discusses BigGAN, an approach that scales up GANs to significantly improve image quality, focusing on large-scale training techniques.

LITERATURE SURVEY NO 4:

Title: "Image-to-Image Translation with Conditional Adversarial Networks"

Authors: Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, Alexei A. Efros

Synopsis: This research introduces a conditional GAN framework, enabling a variety of image-to-image translation tasks, such as photo enhancement and photo-realistic rendering.

LITERATURE SURVEY NO 5:

Title: "Style-Based GAN Architecture for Generative Networks"

Authors: Tero Karras, Samuli Laine, Timo Aila

Synopsis: The authors propose a novel GAN architecture that controls the synthesis process with styles, leading to improved image quality and variety.

LITERATURE SURVEY NO 6:

Title: "GANs for Biological Image Synthesis"

Authors: Tim Sainburg, Marvin Thielk, Brad Theilman, Benjamin Migliori

Synopsis: This project explores the application of GANs in biological image synthesis, demonstrating their potential in scientific imaging.

LITERATURE SURVEY NO 7:

Title: "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks"

Authors: Alec Radford, Luke Metz, Soumith Chintala

Synopsis: This work investigates deep convolutional architectures in GANs for unsupervised learning, offering significant advancements in image synthesis quality.

LITERATURE SURVEY NO 8:

Title: "Cycle GAN: Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks"

Authors: Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros

Synopsis: The paper introduces Cycle GAN, a technique for translating images from one domain to another without paired examples, useful in various image transformation tasks.

CHAPTER 4

EXISTING SYSTEM

The existing intruder alarm systems rely heavily on traditional methods, employing wired sensors and standalone alarms. While these systems provide a certain level of security, they come with several limitations. Firstly, their reliance on wired sensors can restrict their effectiveness and adaptability, as they may not be easily reconfigured or expanded to cover new areas. Additionally, these systems often lack remote monitoring capabilities, making it challenging for users to monitor their property from afar, limiting their ability to respond promptly to security breaches.

Another drawback of traditional intruder alarm systems is their susceptibility to false alarms. These systems often rely on simple detection mechanisms, which may trigger false alarms due to non-threatening stimuli such as pets or environmental factors. Furthermore, the installation process for wired intruder alarm systems can be complex, often requiring professional assistance for proper setup. This complexity adds to the initial cost and may deter some users from adopting these systems.

Maintenance and upgrades for traditional intruder alarm systems can also pose challenges. Over time, these systems may require maintenance to ensure their continued functionality, and upgrades may be necessary to address evolving security threats. However, navigating the maintenance and upgrade process can be daunting for users, potentially leading to increased costs and hindering the system's effectiveness in providing reliable security.

CHAPTER 5

PROPOSED SYSTEM

The proposed IoT-based intruder alarm system represents a significant advancement in security technology, addressing the limitations of traditional systems while leveraging modern IoT capabilities for enhanced functionality and user convenience.

The core of the proposed system revolves around the ESP32 microcontroller, a versatile and powerful platform known for its Wi-Fi connectivity and processing capabilities. Integrated with Passive Infrared (PIR) sensors, the system can accurately detect motion within the monitored area. Upon detecting any suspicious activity, the PIR sensors trigger an alert signal to the ESP32, initiating the alarm sequence.

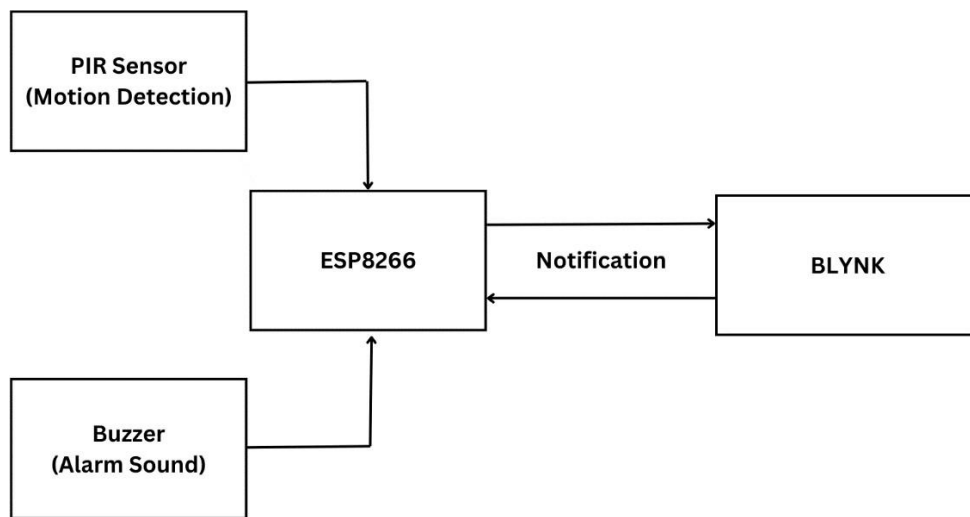
To provide immediate notification to users, a buzzer is integrated into the system. When an intrusion is detected, the buzzer emits a loud sound, alerting occupants and deterring potential intruders. Additionally, the ESP32's Wi-Fi connectivity enables seamless communication with a mobile application, allowing users to monitor and control the alarm system remotely.

Through the mobile application, users can receive real-time alerts, view the status of the alarm system, and even arm or disarm the system as needed. This remote-control capability enhances user convenience and flexibility, enabling users to manage their security system from anywhere with an internet connection.

Furthermore, the proposed system offers easy installation and maintenance. With wireless connectivity and simplified setup procedures, users can deploy the system quickly without the need for professional assistance. Additionally, the modular design allows for scalability, enabling users to expand the system to cover larger areas or integrate additional sensors for comprehensive security coverage.

CHAPTER 6

BLOCL DIAGRAM



CHAPTER 7

HARDWARE REQUIREMENTS

- ESP8266
- PIR Sensor
- Buzzer
- LED

7.1. ESP8266

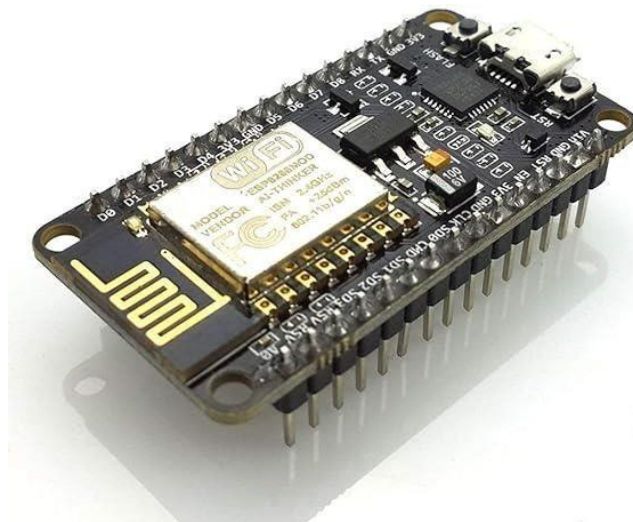


Fig. 7.1 ESP8622

The ESP8266 is a low-cost Wi-Fi microcontroller, with built-in TCP/IP networking software, and microcontroller capability, produced by Espressif Systems in Shanghai, China.

The chip was popularized in the English-speaking maker community in August 2014 via the ESP-01 module, made by a third-party manufacturer Ai-Thinker. This small module allows microcontrollers to connect to a Wi-Fi network and make simple TCP/IP connections using Hayes-style commands. However, at first, there was almost no English-language documentation on the chip and the commands it accepted. The very low price and the fact that there were very few external components on the module, which suggested that it could eventually be very inexpensive in volume, attracted many hackers to explore the module, the chip, and the software on it, as well as to translate the Chinese documentation.

Features:

- **ESP8266** is a Processor: L106 32-bit RISC microprocessor core based on the Tensilica Diamond Standard 106Micro running at 80 or 160 MHz
- **Memory**
 - 32 KiB instruction RAM
 - 32 KiB instruction cache RAM
 - 80 KiB user-data RAM
 - 16 KiB ETS system-data RAM
- **External QSPI flash:** up to 16 MiB is supported (512 KiB to 4 MiB typically included)
- IEEE 802.11 b/g/n Wi-Fi
- Integrated TR switch, balun, LNA, power amplifier and matching network
- WEP or WPA/WPA2 authentication, or open networks
- 17 GPIO pins
- Serial Peripheral Interface Bus (SPI)
- I²C (software implementation)
- I²S interfaces with DMA (sharing pins with GPIO)
- UART on dedicated pins, plus a transmit-only UART can be enabled on GPIO2

Following pins are used in ESP8622

1. GND, Ground (0 V)
2. GPIO 2, General-purpose input/output No. 2
3. GPIO 0, General-purpose input/output No. 0
4. RX, Receive data in, also GPIO3
5. VCC, Voltage (+3.3 V; can handle up to 3.6 V)
6. RST, Reset
7. CH_PD, Chip power-down
8. TX, Transmit data out, also GPIO1

7.2. PIR Sensor:



Fig. 7.2 PIR Sensor

A Passive Infrared (PIR) sensor is a device used to detect motion by measuring the infrared (IR) light radiating from objects in its field of view. PIR sensors are commonly used in security systems, automatic lighting controls, and various applications where motion detection is required.

The sensor itself is divided into two halves, each of which detects infrared radiation. When there is no movement, the two halves detect the same amount of IR radiation. When an object moves, it first passes through one half, causing a change in the detected IR levels, and then the other half, creating a differential signal that the sensor interprets as motion.

Features of PIR Sensors:

1. **Wide Detection Range:** Typically can detect motion within a 10-15 meter range.
2. **Wide Field of View:** Often around 110-180 degrees, depending on the design of the Fresnel lens.
3. **Low Power Consumption:** Ideal for battery-operated devices.
4. **Digital or Analog Output:** Can interface with microcontrollers, alarms, lights, or other systems.
5. **Adjustable Sensitivity and Time Delay:** Allows customization of the detection range and duration of the output signal.

Applications:

1. **Security Systems:** Used in motion detectors for alarms and surveillance.
2. **Automatic Lighting:** Turns lights on when motion is detected and off when no motion is present.
3. **Energy Management:** Helps in reducing energy consumption by ensuring lights and systems are only active when needed.
4. **Consumer Electronics:** In devices like automatic door openers, hand dryers, and more.
5. **Smart Home Devices:** Integrated into smart home systems for automating tasks based on presence detection.

Advantages:

1. **Cost-Effective:** PIR sensors are relatively inexpensive.
2. **Reliable:** High reliability in detecting motion of warm objects like humans and animals.
3. **Low Power:** Consumes very little power, making it suitable for battery-operated systems.
4. **Simple Integration:** Easy to interface with other electronic systems and microcontrollers.

Disadvantages:

1. **Limited to Line-of-Sight:** Cannot detect motion through walls or obstacles.
2. **Sensitive to Environmental Conditions:** Can be affected by temperature changes and direct sunlight.
3. **False Alarms:** May trigger false alarms due to pets or other non-relevant movements.

7.3. BUZZER:



Fig. 7.3 Buzzer

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.

The pin configuration of the buzzer is shown below. It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-' symbol or short terminal and it is connected to the GND terminal.

7.4. LED:



Fig. 7.4 LED

An LED (Light Emitting Diode) is a semiconductor device that emits light when an electric current passes through it. LEDs are widely used in various applications due to their efficiency, long lifespan, and versatility. They are available in different colors, sizes, and intensities, making them suitable for a broad range of uses, from indicator lights to sophisticated display and lighting systems.

CHAPTER 8

SOFTWARE REQUIREMENTS

- ARDUINO IDE
- BLYNK

8.1. ARDUINO IDE:



Fig 8.1 Arduino IDE

Arduino IDE is an open-source software, designed by Arduino.cc and mainly used for writing, compiling & uploading code to almost all Arduino Modules. It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process. It is available for all operating systems i.e. MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role in debugging, editing and compiling the code.

A range of Arduino modules available including Arduino Uno, Arduino Mega, Arduino Leonardo, Arduino Micro and many more. Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code. The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board.

The IDE environment mainly contains two basic parts: Editor and Compiler where former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module. This environment supports both C and C++ language.

8.2. BLYNK:



Fig 8.2 Blynk

Blynk is an IoT platform that simplifies the process of creating connected applications. It comprises a mobile app, Blynk server, and libraries that support various hardware like Arduino, Raspberry Pi, and ESP8266. The Blynk app allows users to design custom dashboards using widgets such as buttons, sliders, and graphs to control and monitor devices. Blynk servers manage the communication between the app and the hardware, ensuring seamless interaction.

KEY FEATURES:

- Real-time data logging
- Customizable notifications
- Automation capabilities for setting rules based on conditions or schedules
- Device management features
- Remote monitoring of device status
- Remote firmware updates
- User-friendly interface
- Versatile platform suitable for hobbyists, educators, and professionals
- Supports a wide range of hardware (Arduino, Raspberry Pi, ESP8266, etc.)
- Easy creation and deployment of IoT solutions

CHAPTER 9

WORKING PRINCIPLE

The intruder alert system leveraging an ESP8266, PIR sensor, Telegram bot, and buzzer operates through a series of coordinated steps. Initially, the PIR sensor detects motion within its range and promptly relays this information to the ESP8266 microcontroller. Upon receiving the signal, the ESP8266 swiftly processes the input and activates the buzzer to emit an immediate audible alert, providing a tangible indication of the detected motion. Simultaneously, the ESP8266 establishes a connection to a designated Wi-Fi network, facilitating internet access for subsequent communication tasks. Leveraging this connectivity.

The ESP8266 communicates with a Telegram bot via the Telegram API, conveying a message or notification regarding the detected motion. Hosted on Telegram servers, the bot swiftly forwards this notification to the user's Telegram app or account, ensuring prompt awareness of the potential intrusion. Upon receiving the notification, the user can promptly respond by assessing the situation and taking necessary security measures. This project amalgamates sensor technology, microcontroller functionality, internet connectivity, and instant messaging capabilities to provide a comprehensive intruder alert system capable of real-time detection and remote notification, thereby enhancing security and peace of mind for users.

The intruder alert system, utilizing an ESP8266 microcontroller, PIR sensor, Telegram bot, and buzzer, orchestrates a seamless flow of operations to ensure effective detection and notification of potential intrusions. At its core, the PIR sensor serves as the frontline detector, promptly sensing any motion within its designated coverage area. Upon detection, it promptly dispatches a signal to the ESP8266 microcontroller, which serves as the central processing unit of the system. Here, the ESP8266 swiftly interprets the incoming signal and triggers the buzzer to emit an audible alarm, providing an immediate alert of the detected motion.

Simultaneously, the ESP8266 leverages its connectivity capabilities to establish a connection to a predefined Wi-Fi network, thereby gaining access to the internet. This connectivity proves pivotal as the microcontroller proceeds to communicate with a designated Telegram bot through the Telegram API. By relaying a message or notification detailing the detected motion, the ESP8266 ensures that the Telegram bot remains informed in real-time.

Hosted on Telegram servers, the bot acts as a conduit, promptly relaying the notification to the user's Telegram application or account. This seamless transmission

ensures that the user receives timely awareness of the potential intrusion, empowering them to respond swiftly and appropriately. Whether it involves remotely assessing the situation or initiating immediate security measures, the user benefits from the system's ability to provide instant updates and notifications.

In essence, the integration of sensor technology, microcontroller functionality, internet connectivity, and instant messaging capabilities culminates in a robust intruder alert system. This system not only detects potential threats but also ensures rapid dissemination of information, thereby bolstering security measures and fostering peace of mind for users.

CHAPTER 10

SOURCE CODE

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <ESP8266WiFi.h>

#include <WiFiClientSecure.h>

#include <UniversalTelegramBot.h>

#include <ArduinoJson.h>

// Define pins for ESP8266

#define SDA_PIN 4 // D2

#define SCL_PIN 5 // D1

LiquidCrystal_I2C lcd(0x27, 16, 2);

const char* ssid = "1";

const char* password = "00000000";

#define BOTtoken "7000183111:AAFgnfEam3O5Rt0FhteHBdWm8r8R303egNs"

#define CHAT_ID "1372543735"

// Update the pin numbers to valid GPIO numbers for ESP8266

#define Sensor 14 // D5

#define LEDR 12 // D6

#define LEDG 13 // D7

#define Buzzer 16 // D0

X509List cert(TELEGRAM_CERTIFICATE_ROOT);
```

```

WiFiClientSecure client;

UniversalTelegramBot bot(BOTtoken, client);

void setup() {

  Serial.begin(115200);

  lcd.init();

  lcd.backlight();

  Wire.begin(SDA_PIN, SCL_PIN); // Initialize I2C with correct pins

  configTime(0, 0, "pool.ntp.org"); // get UTC time via NTP

  client.setTrustAnchors(&cert); // Add root certificate for api.telegram.org

  pinMode(Sensor, INPUT);

  pinMode(LED_R, OUTPUT);

  pinMode(LED_G, OUTPUT);

  pinMode(Buzzer, OUTPUT);

  WiFi.mode(WIFI_STA);

  WiFi.begin(ssid, password);

  int a = 0;

  while (WiFi.status() != WL_CONNECTED) {

    Serial.print(".");

    lcd.setCursor(a, 0);

    lcd.print(".");

    delay(500);

    a++;

  }

```

```

Serial.println("");

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("WiFi connected");

Serial.println("WiFi connected");

Serial.print("IP address: ");

lcd.setCursor(0, 1);

lcd.print(WiFi.localIP());

Serial.println(WiFi.localIP());

delay(500);

bot.sendMessage(CHAT_ID, "System started", "");

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("System started");

delay(1000);

lcd.clear();

}

void loop() {

    bool value = digitalRead(Sensor);

    Serial.print("Sensor value: ");

    Serial.println(value);

    if (value == HIGH) {

```

```
Serial.println("Motion Detected");

digitalWrite(LED1, HIGH);

digitalWrite(Buzzer, HIGH);

digitalWrite(LED2, LOW);

lcd.setCursor(0, 0);

lcd.print("Motion Detected");

bot.sendMessage(CHAT_ID, "Motion detected!!", "");

} else {

    digitalWrite(LED1, LOW);

    digitalWrite(Buzzer, LOW);

    digitalWrite(LED2, HIGH);

    lcd.setCursor(0, 0);

    lcd.print("No Motion ");

}

delay(200); // Short delay to stabilize sensor reading

}
```

CHAPTER 11

RESULTS

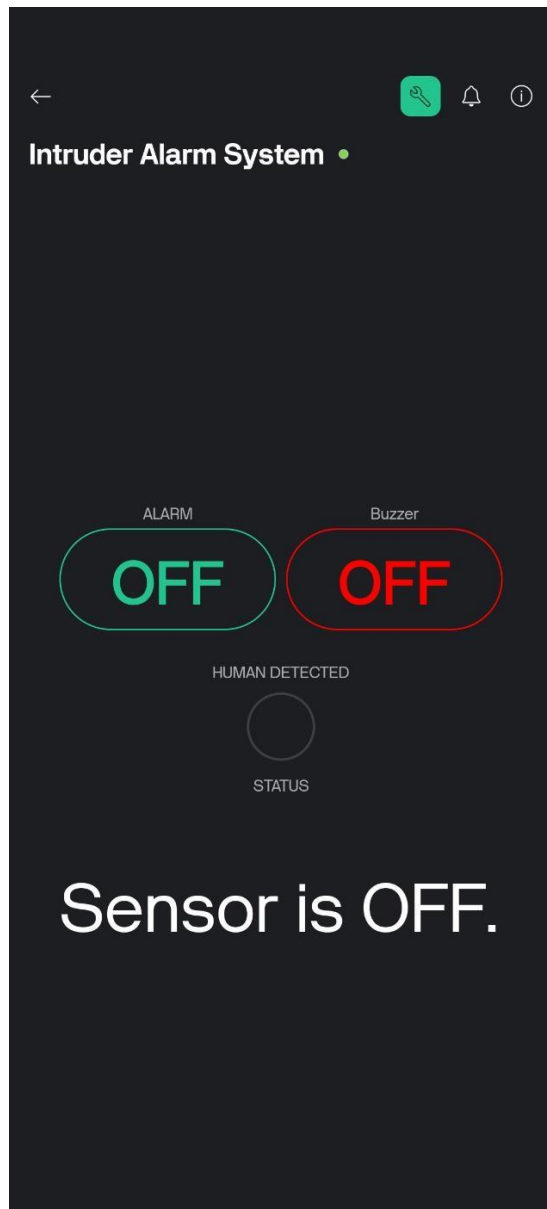


Fig 11.1 OFF State

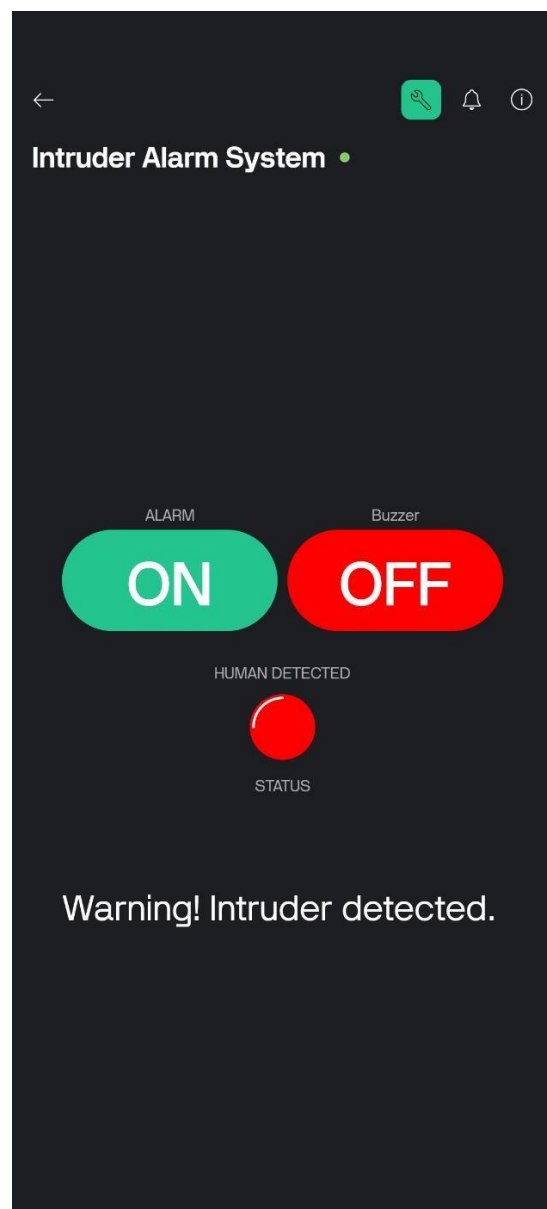


Fig 11.2 Sensor Detected state

CHAPTER 12

CONCLUSION

Integrating IoT technology into alarm systems represents a significant advancement in security solutions, providing enhanced real-time monitoring, remote access, and automated alerts. These systems leverage the power of interconnected sensors, microcontrollers, and communication modules to offer comprehensive and efficient protection. The ability to control and monitor alarm systems from anywhere via smartphones ensures continuous vigilance and responsiveness. Moreover, IoT alarm systems seamlessly integrate with other smart home devices, creating a cohesive and automated security network. Despite challenges such as data privacy concerns, cybersecurity threats, and interoperability issues, these can be mitigated with robust encryption, regular updates, and adherence to standard protocols.

The cost-effectiveness of IoT alarm systems is evident in their energy efficiency and reduced maintenance needs, further enhanced by data analytics for predictive maintenance and system optimization. IoT alarm systems utilize data from various sensors to predict potential issues, allowing for pre-emptive measures that enhance system reliability and longevity. This predictive capability also enables customization of security protocols based on user behavior, providing tailored security solutions.

Overall, IoT integration transforms alarm systems into sophisticated, user-friendly, and highly reliable security tools. They offer not just enhanced security but also peace of mind, knowing that one's property and loved ones are protected by the latest technology. The ongoing innovations and improvements in IoT technology will continue to expand the capabilities and applications of these systems, making them an indispensable part of modern security solutions.

CHAPTER 13

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